

Chapter 7, Problem 83P

Problem

An RC circuit consists of a series connection of a 120-V source, a switch, a 34-M Ω resistor, and a 15- μ F capacitor. The circuit is used in estimating the speed of a horse running a 4-km racetrack. The switch closes when the horse begins and opens when the horse crosses the finish line. Assuming that the capacitor charges to 85.6 V, calculate the speed of the horse.

Step-by-step solution

Step 1 of 1

$$V(\infty) = 120.$$

$$V(0) = 0$$

$$\tau = RC = 34 \times 10^6 \times 15 \times 10^{-6} \\ = 510 \text{ S.}$$

$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-\frac{t}{\tau}}$$

$$85.6 = 120 \left[1 - e^{-\frac{t}{510}} \right]$$

Solving for t gives,

$$t = 510 \ln 3.488 = 637.16 \text{ S.}$$

$$\text{Speed} = \frac{4000 \text{ m}}{637.16 \text{ S}}$$

$$= 6.278 \text{ m/S.}$$